CHAPTER 8

FINDING LAP TIME

A Data Collection System can help find the most common areas where lap time is lost and help you formulate a plan for going faster.

ANYONE WHO DRIVES A racecar knows deep in their heart what they really want to hear: "You've got the fastest time by over a second." Whether it's practice, qualifying, or a race, we all live for those precious words. It doesn't happen often.

It's much more common, especially when you are just starting your career, to find that you're full seconds off the fastest pace. Finding this out, drenched in sweat after trying as hard as you can to go as fast as possible, can be dispiriting. You ask yourself, "Where am I going to find (fill in the blank) seconds?"

SEBRING—OUR LABORATORY

In this chapter, we'll look at some laps of Sebring and see where that precious lap time comes from. In the preceding chapter, you saw the level of detail we put into the analysis of the course. Here, the detail is in the speeds and tenths of seconds that separate a slow lap from a lap record. We'll concentrate on laps driven in our Formula Dodge race cars which we use in our Racing Schools and in our Formula Dodge Race Series. Since such a wide variety of drivers compete in these cars at Sebring, we have a huge sample of unique laps on which to draw.

In addition to first-hand feedback, we can rely on our data acquisition system to answer very specific questions about a variety of laps. Computers and the data acquisition systems that feed them have become a commonplace tool in the testing and development of modern racecars. There are a variety of systems in use now: some monitor engine systems, some keep an eye on speeds and RPM, and some look at the G forces that the car is generating around the course.

The MRG/Skip Barber Performance Monitor

The system we use was developed by Chris Wallach of Marblehead Racing Group (MRG). Chris's hardware is a combination of a memory unit, a control system and a variety of transducers which provide the raw data. The raw information is made accessible by the MRG Performance Monitor software package, which produces a variety of reports and user-designed graphs that make the car's performance come alive in a usable, understandable format.

In most race teams, the system is used primarily as a car-sorting tool. Changes can be made to the racecar and the driver can do back to back runs while the performance monitor looks closely at the results of the changes. "Closely" is an understatement—the monitor looks at up to 250 samples of data a second, and the type of information available is limited only by the development engineer's imagination. There are sensors available to measure air pressure changes, suspension movement, tire loading, and even tire temperature.

In looking at driving technique, we're looking for simple things. First, we'd like to know where the racecar is on the circuit so that when an event takes place, like the driver applying the throttle, we can say definitively that the throttle came on 50 feet before the apex, or worse, 50 feet *after* the apex.

We use two systems to accomplish this. The first is a series of infrared beams set up at specific measured distances around the racetrack. A sensor on the racecar recognizes these beams and the system makes note of the elapsed time from beam to beam. The computer gives us a running lap time as well as the times for each segment of the course.

The system also gives us the car's speed by counting the tire or driveshaft revolutions, so by knowing the speed and having frequent updates on the car's position, we can locate the car on any portion of the circuit to an accuracy of +/– 4 feet.

Now for the driving part: we want to know how the driver 1) applies the throttle, 2) uses the brakes, and 3) turns the steering wheel. To get the throttle information, we have a transducer that measures the percentage of throttle application from 100% to 0%. For braking, we're interested in how much pressure the driver exerts on the brake pedal, so we have a transducer that gives us pedal pressure in pounds of force at the pedal. In the steering department, we're able to measure the steering wheel deflection in degrees. In order to see the forces that are created by all this motion of the controls, we have sensitive accelerometers on the car that measure braking, acceleration, and cornering force in Gs.

All of this data is time referenced, meaning that the system takes a snapshot, if you will, of all these different channels of information between 20 to 250 times a second (user's choice) as the car proceeds around the course. By integrating the time and speed, we can create distance-based graphs which show where the car is on the racetrack at any time.

When analyzing a driver's performance, our most useful tool is the primary display which shows brake pressure, throttle position in bar graph style, and the position of the steering wheel, represented by a little steering wheel. The primary screen allows us to display two laps simultaneously—normally a target lap set by an instructor and a lap by the driver being coached. It paints a pretty clear picture of what the driver is doing in the car and how the driver's technique results in gains and losses of lap time.

FROM WARM-UP TO FAST LAPS

"Do a *warm-up* lap or two": a simple statement and one which most of us understand, subjectively. The degree to which a warm-up lap is slower than a race pace depends upon the driver, of course, but let's look at the difference between what an experienced driver does warming up and what he does on the fastest lap of the day.

The fastest flying lap on this particular day in car No. 22 will be a 1: 23.85 lap. The warm-up is a 1:29.4, over 5.5 seconds per lap slower—a huge difference in racing terms, but on a percentage basis, only 6% slower. Let's take a look at the accompanying breakdown of each lap to see where this time comes from.

Note Speed Differences

First, peruse the differences in speed between the warm-up and fast lap at key points around the racetrack (see Fig. 8-1). The first thing that pops out is that the corner exit speeds in the slow corners are not very different in the two laps. This is a key fact to remember: when you're trying to lower lap time, you're usually not looking for a 10 m.p.h. improvement, especially in the corners taken below third gear (3, 6, 7, 9, 10). The difference in corner exit speeds for laps that vary by 5.5 seconds per lap is, *at most*, 3 m.p.h.

All of these corners are first or second gear corners, and the fact that the exit speeds are close has a lot to do with the comfort level of accelerating and turning in the lower gears. You're in control, pushing on the accelerator, and if the path seems wrong, a little breathe out of the throttle fixes things. Also, when you're driving under 70 m.p.h., there's nothing intimidating about using aggressive power coming off of a slow corner. As you work your way to faster laps,

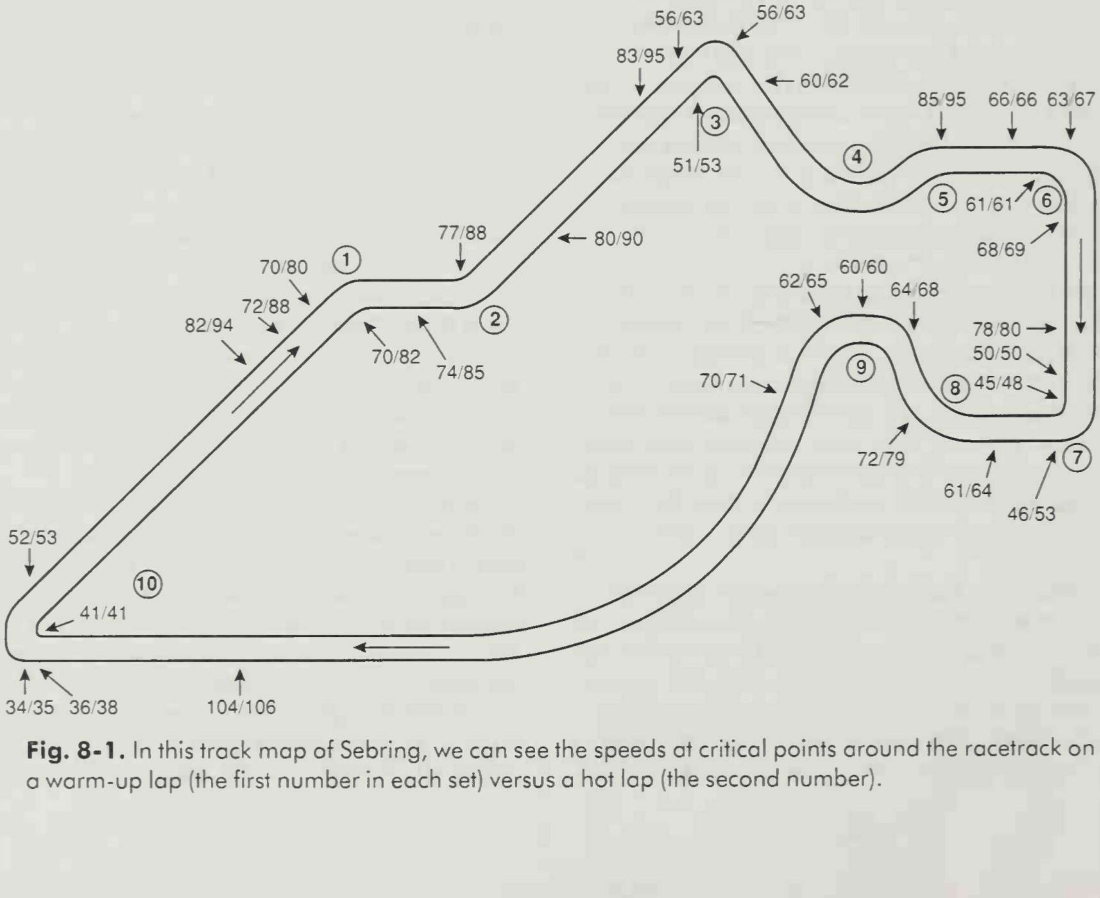

Fig. 8-1. In this track map of Sebring, we can see the speeds at critical points around the racetrack on a warm-up lap (the first number in each set) versus a hot lap (the second number).

don't expect that your exit speed out of slow turns has to increase by a huge amount in order to make up significant time.

Another factor at play here is experience level. Although this is a warm-up, our driver has vast experience at this racetrack—the line through the corners is practically instinctive and, consequently, very accurate. You couldn't say the same if it were a driver's first-ever lap.

Identify the Most Important Corners

This small speed difference coming out of slow corners puts a new wrinkle on the theory of identifying the most important corners on a racetrack. In "The Real-World Line" we pointed out that the more important corners were those which led onto long periods of full throttle acceleration. In practical terms, however, drivers of widely differing skill levels seem to be able to closely match speeds coming out of slow corners; getting a significant advantage over others in the field in these slow corners is rare. A fast corner leading onto a long straight is another matter.

In Turns 1 and 2 the speeds between the warm-up and the fast lap differ by over 10 m.p.h. Partly it's a matter of relative comfort: the high speed stuff is going to be more difficult than a second gear turn. Also, in the specific case of Sebring there is the problem of a scarcity of obvious turn-in reference points for Turns 1 and 2. For the first few laps in a racecar you're still looking for points which will work to guide you to the best line.

This cautiousness in the fast corners costs plenty. In Turns 1 and 2, the segment time for the warm-up is .95 seconds slower. This doesn't count the corner entries, just the time from the apex of 1 to the entry of 3. Almost a full second of our 5.5 second deficit is going to come from driving through 1 and 2 faster.

Faster corners are longer, and the difference in speed between a driver who's right on the limit and one who is just shy lasts for a longer time, producing a bigger difference in lap time. Couple a fast corner with a long straight, and the difference gets magnified even further.

Compare Segment Times

Now, let's figure out how much of the rest of the 5.5 second shortfall is coming out of the corners and driving down the following straights, and how much is in the braking zones.

In the track map in Fig. 8-2 the course is divided into corner exit zones (from the apex to the next corner entry) and corner entry zones (from the braking point to the apex). As you can see from the table in Fig. 8-3, there is a 3.2 second difference in lap time in the corner exit segments and a 2.35 second difference in the corner entry splits. This really shouldn't come as a surprise, since there is a lot more distance covered and time spent on the corner exit portion of the course. What is important to remember is that corner exit speed is easier to get than corner entry speed.

You can see why we stress using a progression of 1) concentrating on line, 2) working on getting greater

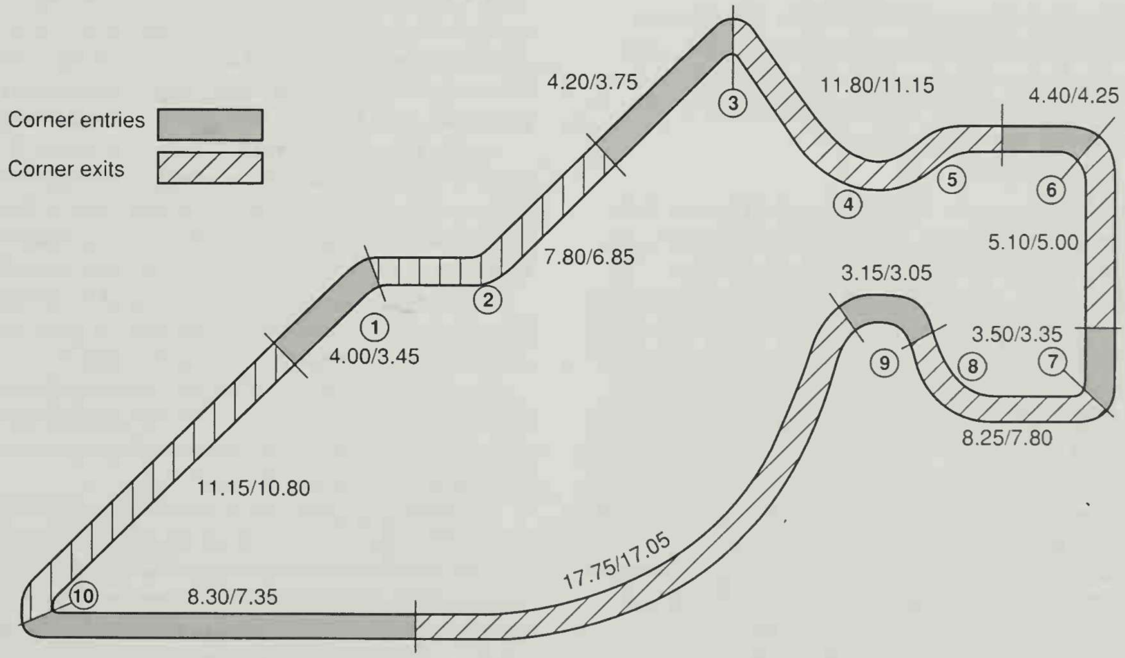

Fig. 8-2. In this map of the Sebring course, compare the corner entry and corner exit segment time differences in a warm-up vs. a fast lap (first number in set is warm-up time, second number is hot lap time). There is significant time to be gained by concentrating on corner exits before approaching the limits on corner entry.

Corner Exits	Warm-up Lap	Fast Lap	Time Shortfall
Exiting Turn 10 to Turn 1	11.15	10.80	.35
Exiting Turn 1 to Turn 3	7.80	6.85	.95
Exiting Turn 3 to Turn 6	11.80	11.15	.65
Exiting Turn 6 to Turn 7	5.10	5.00	.10
Exiting Turn 7 to Turn 9	8.25	7.80	.45
Exiting Turn 9 to Turn 10	17.75	17.05	.70
Total (sec.)	61.85	58.65	3.20

Corner Entries	Warm-up Lap	Fast Lap	Time Shortfall
Entering Turn 1	4.00	3.45	.55
Entering Turn 3	4.20	3.75	.45
Entering Turn 6	4.40	4.25	.15
Entering Turn 7	3.50	3.35	.15
Entering Turn 9	3.15	3.05	.10
Entering Turn 10	8.30	7.35	.95
Total (sec.)	27.55	25.20	2.35

Fig. 8-3. These tables reveal the important corner exits and entries at Sebring for gaining lap time (warm-up lap vs. fast lap). Taken separately, the time losses may look like small potatoes, but they can really add up.

cornering speed and exit speed, and 3) finally working on optimizing corner entries. In this Sebring example you wouldn't worry about using the car very hard until you came close to consistently putting the car in the right place. If you did just this, you could get within 5.55 seconds of the fastest time. Increase exit speed in the slow corners and use full acceleration and crisp upshifting, and you can knock off 1.35 seconds just on the two longest straights (3 to 6 and 9 to 10). Add in all the other acceleration zones, and there is another .90 seconds available. Pick up the pace in 1 and 2, and you get almost another full second.

Without concentrating on braking and entry speed you could do a 1:26.2 versus a fastest of the day at 1:23.85. That's not very far off, considering that you've paid no attention to trying to brake late or carry speed into the corners.

Corner Exit: First Place to Make Up Lap Time

Once warm-ups are over, you can make up this time in two ways. The first is running closer to the car's limit coming away from the two important corners which lead onto the longest straights. It's interesting to see just how little exit speed improvement is needed.The second chunk of the 2.25 seconds available in the acceleration zones comes from using full power and minimizing the time between gears. In the warm-up, upshifts took close to a half second. In the fast lap, the throttle pedal went from full on before the shift, to off between gears, and back to full on in just under two tenths of a second. The three tenths difference at each shift adds up to straightaway speed and lap time.

The catch to faster shifting is that the more you rush it, the greater the chance of missing a gear and over-revving the motor. You can be conservative and take a bit more time with your upshifts, but it will cost you lap time. You have to develop the ability to coordinate your feet and your right hand to change up quickly and accurately.

The .95 second difference in 1 and 2 is going to be harder because of the difficulties we noted earlier, but the 10 m.p.h. speed deficit can be closed by taking ten 1-m.p.h. nibbles, working the comfort level up, without doing anything dramatic at the corner entry. Ten 1-m.p.h. nibbles get you into contention: two 5-m.p.h. bites get you in big trouble.

Corner Entries: Last Bit of Time

Using the map in Fig. 8-2 and the table in Fig. 8-3, let's look at the shortfall on corner entries. Starting with Turn 1, the time is off .55 seconds from the lap record—not surprising since the turn-in speed is 10 m.p.h. slower than the fast lap. The braking in the warm-up is long and light and just picking up the pace, even with light braking, will make up tenths easily.

At the approach to Turn 3 there is another .45 seconds: a 12 m.p.h. difference in approach speed and 7 m.p.h. at the turn-in. Pick up the pace out of Turn 2 and, without braking any harder, carrying some more speed up to the turn-in will make up half the deficit.

The corner entries of Turns 6, 7, and 9 together amount to just .4 seconds, so you won't expect to have to change the approach to these very much.

The big corner entry difference is the approach to The Hairpin, Turn 10, with a .95 second difference. There is just as much lap time available here as in picking up the pace in the esses.

In the fast lap, the brakes come on just 280 feet short of the turn-in point while in the warm-up, the braking uses less than half the pedal pressure and starts almost 200 feet earlier. The message here is that the high speed-loss corners are going to be the ones where approaching the braking limits will have the biggest effect on lap time. Keep in mind, however, that part of this lap time difference comes from the speed to which you slow the car at the turn-in point. In the fast lap, the car is 2 m.p.h. faster at the turn-in point.

Consider that on the warm-up, you're giving away a total of 1.5 seconds on the corner entry of 5 other corners and almost a full second approaching The Hairpin. This information should lead you to another conclusion about the order of efforts to reduce lap time: not only should you leave limit braking for last, but once you concentrate on braking, there is much more to be gained by starting with the high speed-loss braking zones.

CHASING A FASTER PACE

In the warm-up comparison, one driver is choosing between one pace and another. The more important comparison is between a driver who is at the lap record pace and one who can't get there, even though he or she is trying hard. Let's use the data collection system to see if we can identify common sources of problems with drivers who have the basics covered but find themselves off the pace.

Computer Coaching

We'll use the same techniques we use in our "Computer Lapping" program at Skip Barber Racing School. An instructor shakes down the car first thing in the morning to set a target lap for comparison purposes. It's important that this target lap be done in the same car on the same day as the driver who's being coached. As we said before, conditions change day to day, and we want to be certain that we're making apples to apples comparisons. On the day we'll be looking at, our data collection instructor, Bruce McQuiston, puts in a real flyer—a 1:23.60—the fastest lap yet.

Within the hour, the driver we'll be coaching does 20 minutes of warm-up (he'd done 80 miles of testing the day before), then a six-lap collection of data. The fastest lap of his collection is a 1:25.55, within two seconds of the target. At first, two seconds sounds like a huge difference, but as you work your way around the lap (see Fig. 8-4) you can see that there aren't any glaring exit speed or straightaway speed differences. There is a mile per hour here or there but you wouldn't think that there is two seconds worth.

You can see, however, that the largest speed differences seem to come at the corner entries, especially at the minimum speed points, which are found between the turn-in and the apex. While these speed comparisons are interesting, what really helps the pursuit of lower lap time is to look at the segment times.

The first step in the analysis is to compare the segment times of each collection. The data collection software allows us to lay one lap over another and scroll through each segment of the track to see the segment time differences in each section (Fig. 8-5).

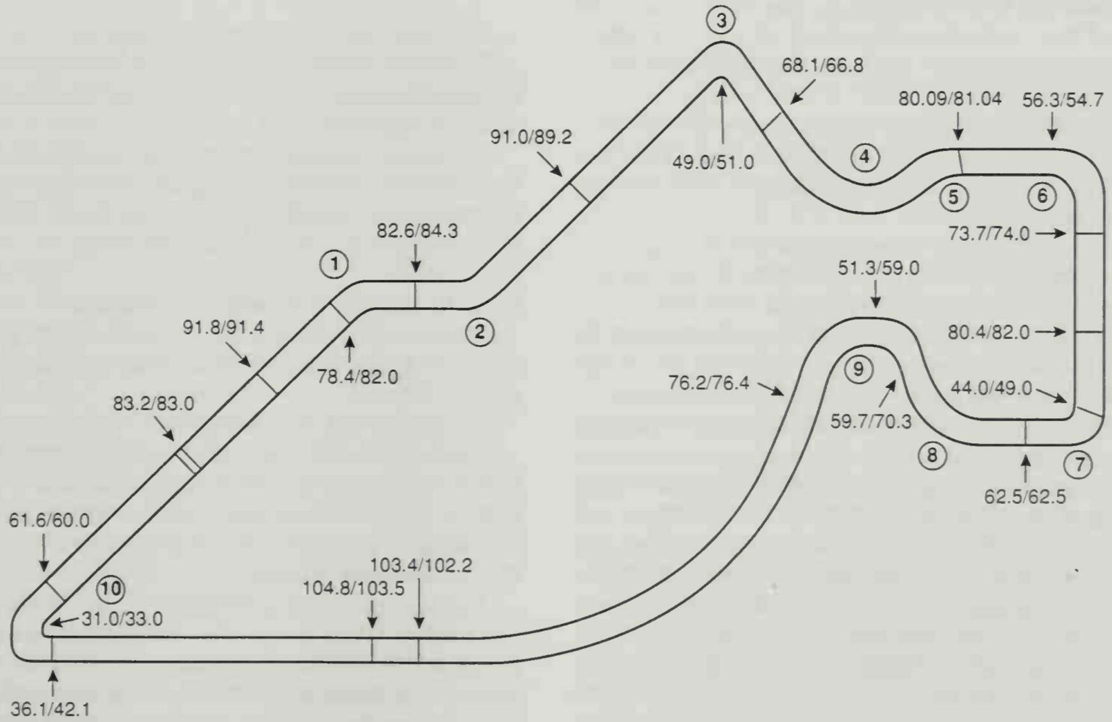

Fig. 8-4. This track map compares the speeds of the two drivers—the instructor and the client—at significant points around the course (first number in each set is client speed in m.p.h., second is target speed). The largest differences in speed seem to come at the corner entries.

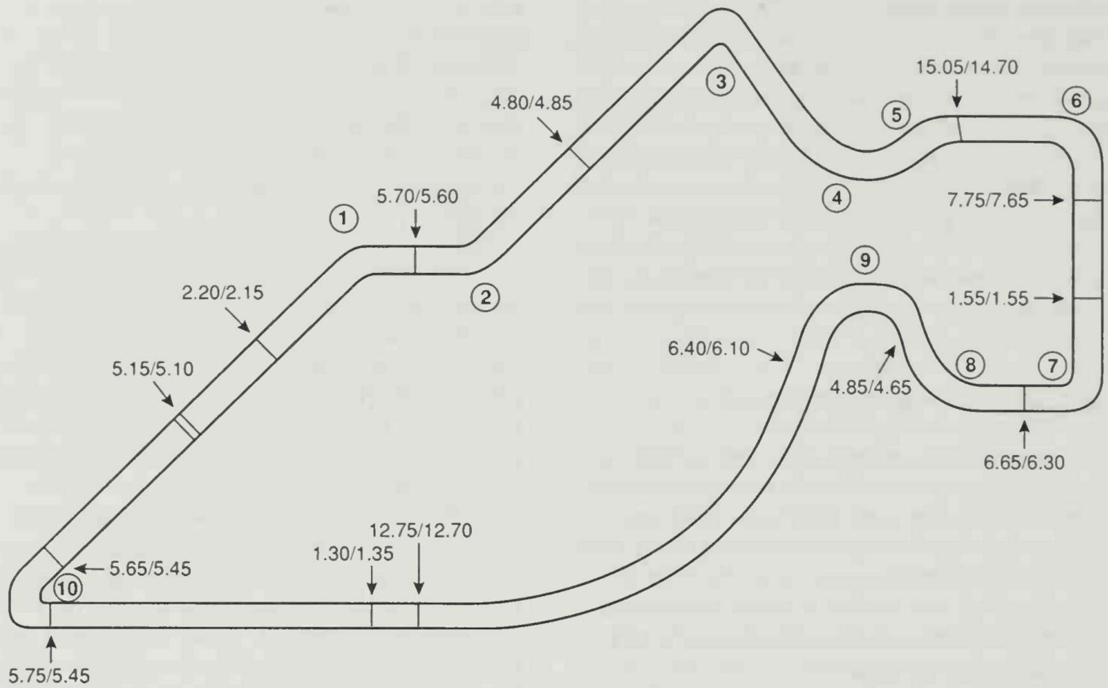

Fig. 8-5. This track map highlights the differences in segment times between the two laps (measured in seconds; first number in each set is client time, second is instructor time). Concentrating on the corners with the largest segment time differences allows you to focus on problem areas.

Looking for the Biggest Shortfall

Beginning at Start/Finish and going right through the exit of Turn 2, there is only a tenth of a second difference between the two runs—no big problem in the fast stuff.

The next big segment, from the approach to Turn 3 up to the entry of Turn 6, is a different story. The slower lap shows a loss of .35 seconds here, definitely worth looking at.

Moving along, the spread in the turn six segment and up to turn seven is another wash—a tenth of a second over a nine second segment.

Big differences begin to show up between the entry of turn seven and the exit of turn nine. The slower lap is off .35 seconds just in Turn 7, loses another two tenths up to Turn 9 and another three tenths in 9 itself. This relatively small section of the racetrack accounts for .85 seconds of the lap time deficit. You have to dig a little deeper to find out why.

The main straight is even, both segment times are identical, but there is a real difference, three tenths, in the deceleration zone for the hairpin. Another two tenths is available in the hairpin itself, so you're looking for a half-second on the approach to, and the run through, The Hairpin.

WHERE THE TIME GOES

Let's go after the biggest chunk of lap time first, the entry to Turn 7 to the exit of Turn 9, looking at the speed differences through the segment, then the cause of the speed deficit.

You can get the details of the lap by graphing any of the channels of information which are recorded. We usually start with speed, pinpointing where in any segment, the car is slower than the target.

How to Read the Computer Graphs

To make the following graphs more clear, let's go over the main features of the performance monitor primary screen first.

Looking at Fig. 8-6, on the lower left (A) are the particular runs—in this case, the target, NSMQ600, and the session by the driver being coached, NSHW52. To the immediate right are the individual laps we are comparing (B)— the number of the lap in the run and the lap time.

The far right column (C) lists the recorded data. We can choose to graph speed, brake effort, steering angle, lateral G, and longitudinal G (in this series of collections the longitudinal G accelerometer was frozen at a bogus setting, so throughout the following graphs disregard LONGG).

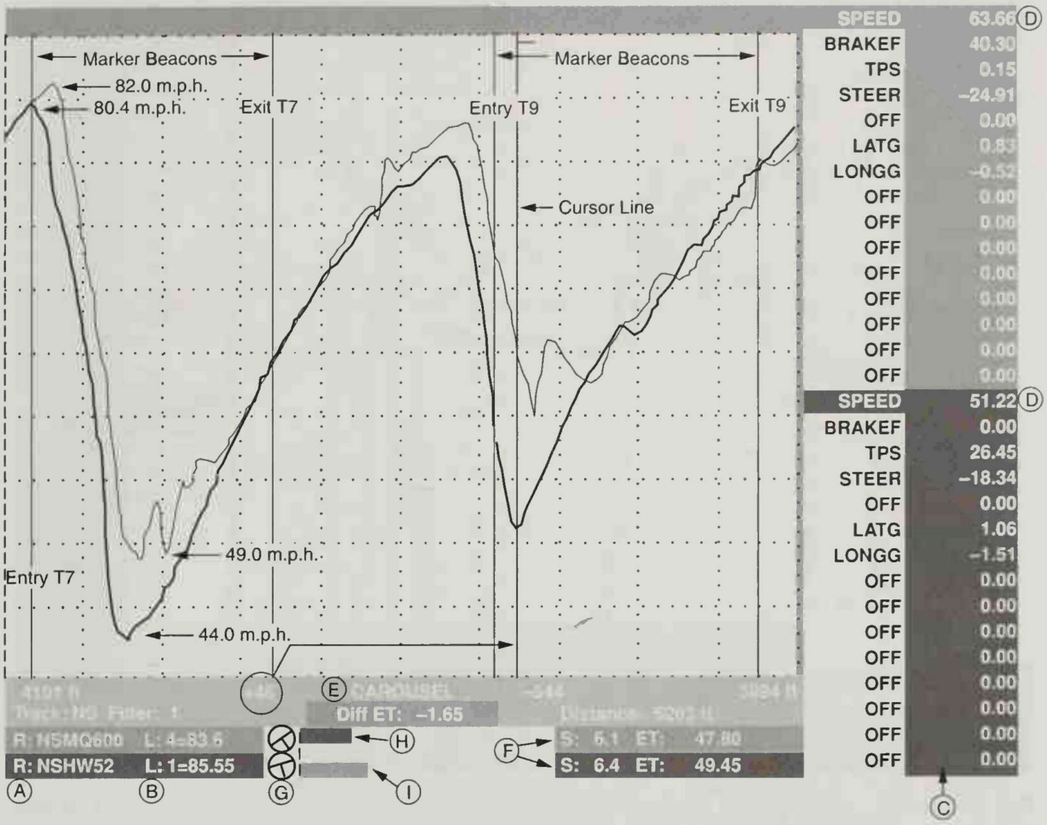

Fig. 8-6. This printout of the primary analysis screen on the performance monitor shows the different aspects of the Turn 7 through Turn 9 segment that can be measured—in this case, speed.

By using the mouse, you can highlight which channel to graph. Here, we have selected just speed (D) for each run. On the graph itself there are a series of vertical lines. The lighter lines represent the infrared markers defining the beginning and end of a segment of the racetrack. The darker vertical line is a movable cursor line that moves right and left. In this graph this line is placed 46 feet into the "carousel" segment. The tables to the right display the values for each channel of information for wherever the cursor line is placed. In this graph we placed the cursor line at the minimum speed point in the carousel for the slower lap and you can see that the speed table at this point of the lap reads 63.66 m.p.h. for the faster segment vs. 51.22 m.p.h. for the slower pass. As you move the cursor line, the segment it's on is identified (E) and the segment time for the laps being compared is displayed (F). You have the ability to zoom in and out to look at an entire lap or a small segment of the lap.

You can also see how much the steering wheel is turned (G) at the location of the cursor line as well as the relative level of braking force (H) and the level of throttle application (I).

Speed Differences

By just graphing the speeds from the entry of Turn 7 to the exit of Turn 9 (as we have in Fig. 8-6), you can figure out that the problem is not the speed at the approach to the corner, nor is it, for that matter, the corner exit speeds. The main difficulty is that the slower driver is giving away too much speed in the first half of the corner by over-slowing the car.

The first third of the speed graph is the entry through the exit of Turn 7. The slowest speed in the corner on the fast lap is 49 m.p.h., compared to 44 m.p.h. for the slower lap—a 5 m.p.h. spread. This might be caused by braking too soon or too long.

Braking Variations

If you graph brake pedal effort in the two runs (see Fig. 8-7), you can see that the brake point for the slower run was 48 feet earlier than in the target lap. In addition, while the initial level of braking was good, over seventy pounds of brake pressure was released to accommodate the blip for the downshift. It's not a big surprise that using between 40% to 60% braking over a distance of close to 100 feet on the approach to Turn 7 requires that the process start sooner.

We suspected in our first look at the racetrack that this corner, because it is so short, would require rela-

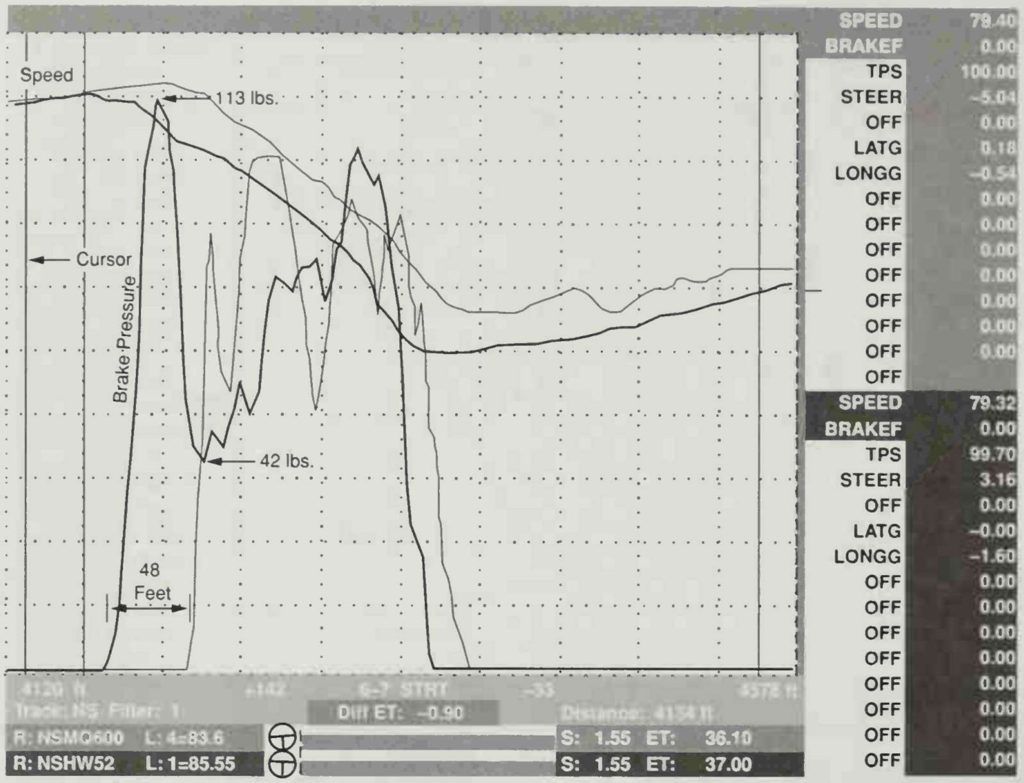

Fig. 8-7. In this graph of both speed and brake pressure for the two runs, in the slower run (darker trace) the brakes come on early, making the speed lower throughout the braking zone. Also, brake pressure is drastically reduced from threshold when the driver blips the throttle for the downshift.

tively little trail-braking, and you can see that this is supported by the data. The faster driver trail-braked just 35 feet past the turn-in and gave a quick application of lots of throttle. The slower driver released the brakes sooner and went back to partial throttle earlier. In the target lap, the throttle application started later, but got to full sooner. However, a breathe off the throttle was necessary in the fast lap to control oversteer. Both drivers were at the same speed, 62.5 m.p.h., at the corner exit beacon.

Not surprisingly, you can see the same overslowing problem at the entry to Turn 9. Since both drivers exited Turn 7 at the same speed, the two tenths segment time difference again comes at the corner entry.

In Fig. 8-8, we've graphed the braking effort and the speed trace for each run from the exit of 7 through the exit of the carousel. The difference in speed is very graphic, a minimum speed of 59 m.p.h. for the target, and 51.3 for the slower lap. You can see that the brake point for the slower run is 40 feet earlier than the target. The loss in entry speed actually started in Turn 8, the right hand sweep up to the turn-in for 9. The slower run shifted to third gear earlier and the acceleration rate in the lower RPM range in third starts to fall off relative to the faster run, which stretches second gear to the redline, then ends up at higher RPM in third gear after the later shift.

The Carousel

Taking a closer look at the braking for the carousel (Fig. 8-9), you can see some significant differences which account for the segment time difference. Again, the slower run starts the braking earlier—40 feet—and there is a significant drop in brake pressure when downshifting. Note also the long gradual trail off braking effort as the car turns in to Turn 9.

In the faster run, the driver starts at threshold on the approach, then trails back to a lower, but constant, level of braking up to the brake-throttle transition. We suspected that this would be a workable approach; since there is a long distance between the turn-in and the throttle application point, you can use the early portion of the corner to slow the car while it's turning.

You will notice in Fig. 8-9 that at the cursor line the speed differential is 5 m.p.h. Yet, looking at the throttle position bar, the slower run is at full throttle while the faster driver is at about 80% throttle. This is possible because at the lower speed there is excess traction available, and the driver can use full throttle without overtaxing the tires' grip, making up some of the deficit at the exit of the corner. It's a trade-off. The slower car can accelerate faster from the lower speed (the speed trace catches up with the faster run at the corner exit).

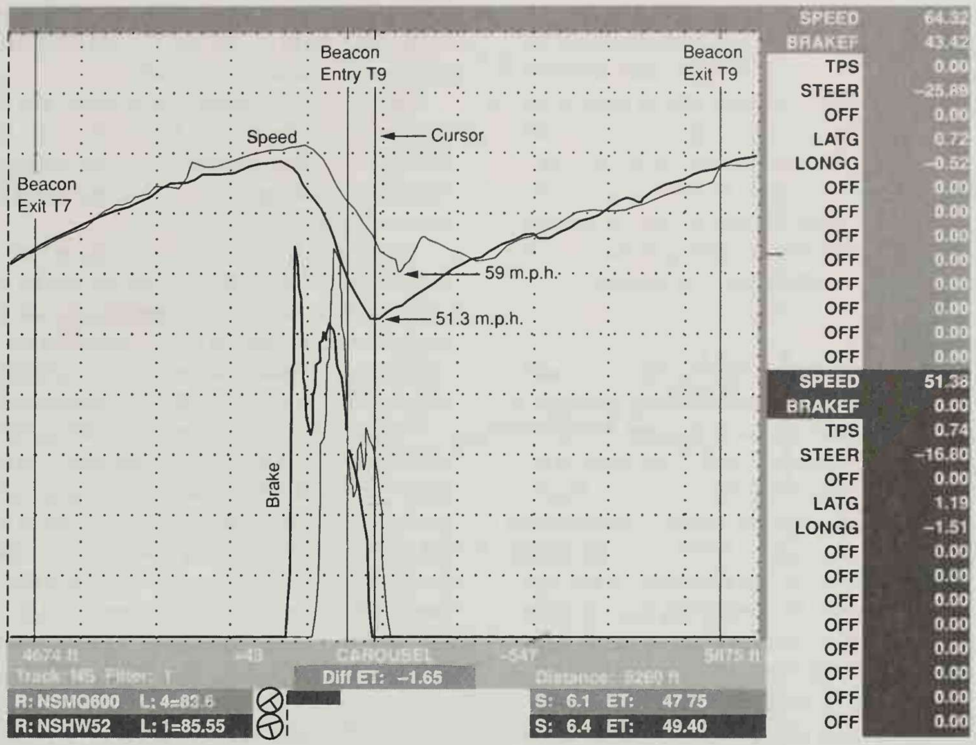

Fig. 8-8. Graphing both speed and brake effort for Turn 7 exit through the carousel exit, you can see that a half-second of lap time is lost, again, on overslowing the car. Note that the speed trace at the exit of both Turn 7 and the carousel are nearly identical.

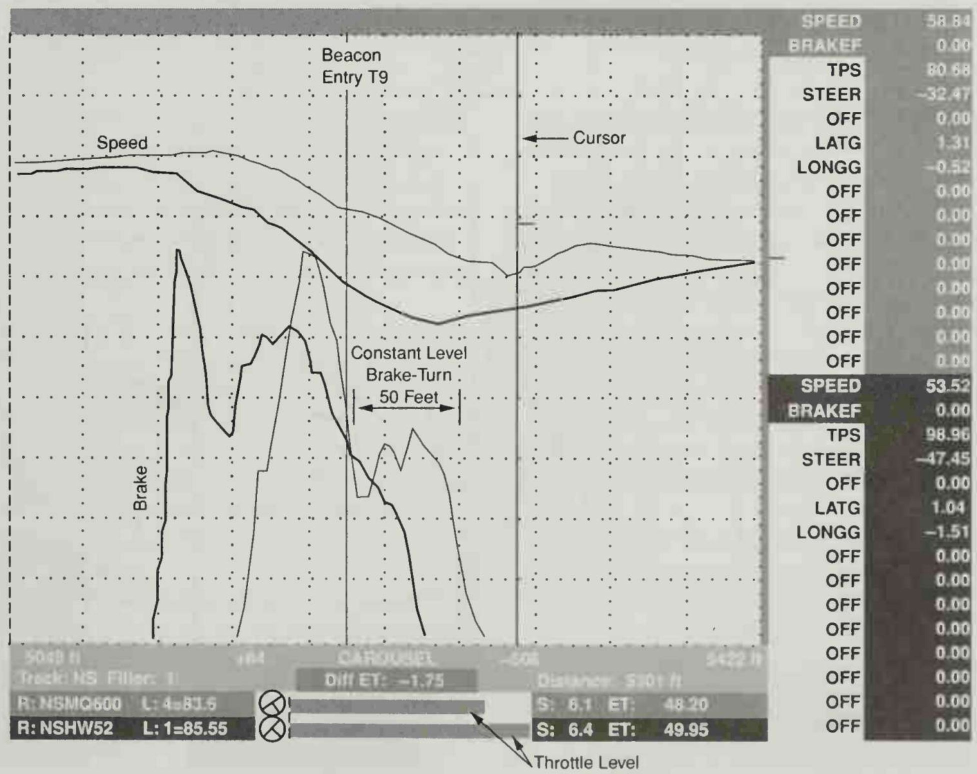

Fig. 8-9. This expanded scale of Fig. 8-8 above shows that the slower run brakes earlier and is 6 m.p.h. slower at the minimum speed point of the corner. The driver was able to use more throttle earlier than the target run, but the superior acceleration didn't make up for the time lost at the corner entry.

In the faster run, since the tires are closer to their traction limits the driver has to be more sensitive to mixing throttle and cornering. Still, acceleration off the corner starts at a higher speed, and as long as exit speed isn't compromised, and a longer segment time down the straight results, the time made up on the corner entry is time in the bank. In this run, no exit speed is lost; and the segment time down the following straight is identical to the car that got the power on sooner, harder, but started 6 m.p.h. slower.

The Hairpin

The next largest chunk of time during the lap, a half-second, comes from the approach through the exit of The Hairpin. Three tenths is lost in the last 700 feet of the straight, up to the turn-in point. If we look at a speed comparison between the two runs (see Fig. 8-10), you can see that the approach speed in the slower run is actually higher but, again, the minimum speed in The Hairpin is 2 m.p.h. lower than the target lap.

The biggest difference, however, lies in how long each driver took to lose the speed. In the faster run the braking starts 281 feet from the turn-in, versus 416 feet for the slower run. The three tenths in this segment comes from driving 135 feet closer to the corner before going to the brakes. Without a doubt, the slower driver wouldn't believe he could go 135 feet deeper, and in fact he couldn't unless he reached a braking level that would allow him to lose the same amount of speed over a shorter distance.

This graph also reveals that there is a big pressure release when the slower driver blips the throttle for the fourth to first downshift, and a long gradual trail out of the braking level on the approach to the corner—not great footwork.

By contrast, the faster driver goes through each gear, three separate blips, and the brake pedal pressure varies by, at most, 20 lbs. The slower driver uses a peak pressure of 84 lbs and a low of 26 lbs, as opposed to the target threshold pressure that varies between 109 and 90 lbs for the bulk of the deceleration.

This is one of those instances when late threshold braking is worth significant lap time. As we said earlier in the chapter on braking, corner approaches where significant speed loss takes place put a premium on threshold braking. Here the premium is a potential three tenths of a second. The underlying lesson here is that just going deeper won't do it—you have to reach closer to the threshold level of pressure *and stay there* despite the blips for the downshifts.

Losing Too Much Speed

If you look again at the speed comparisons at significant parts of the hairpin, you can pick out where the

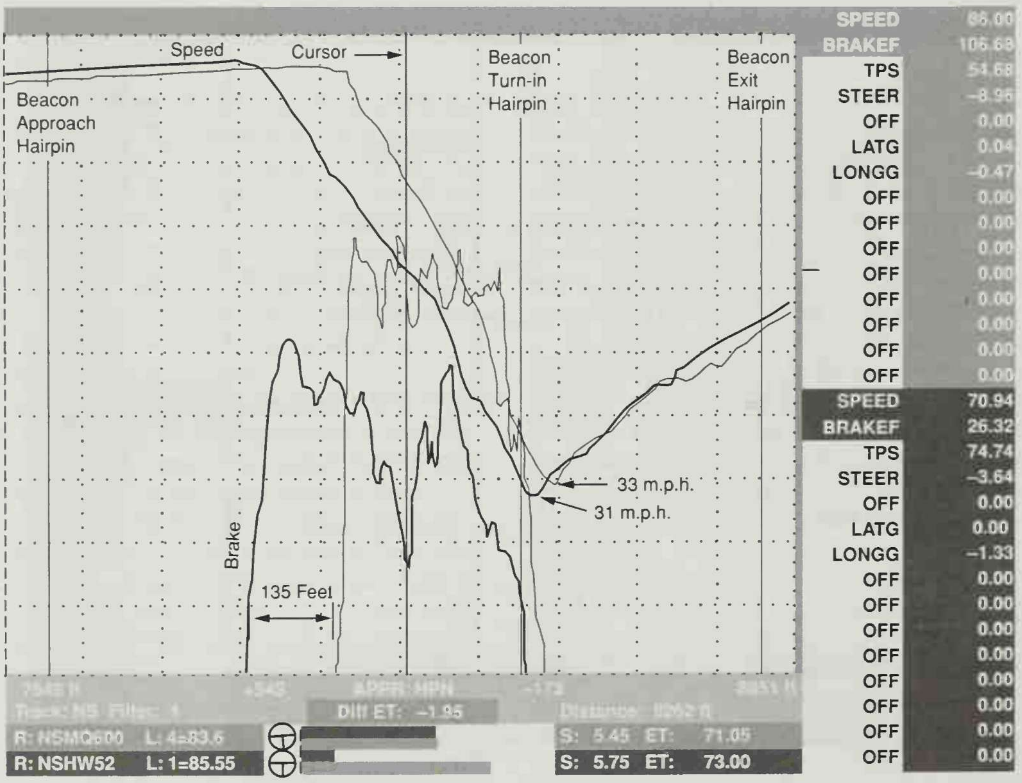

Fig. 8-10. A half-second of lap time is lost on the approach to The Hairpin and The Hairpin itself. You can see on the approach that the problem is light, early braking. In the corner, over-slowing the entry cost .2 seconds even though the exit speed is 1.6 m.p.h. higher.

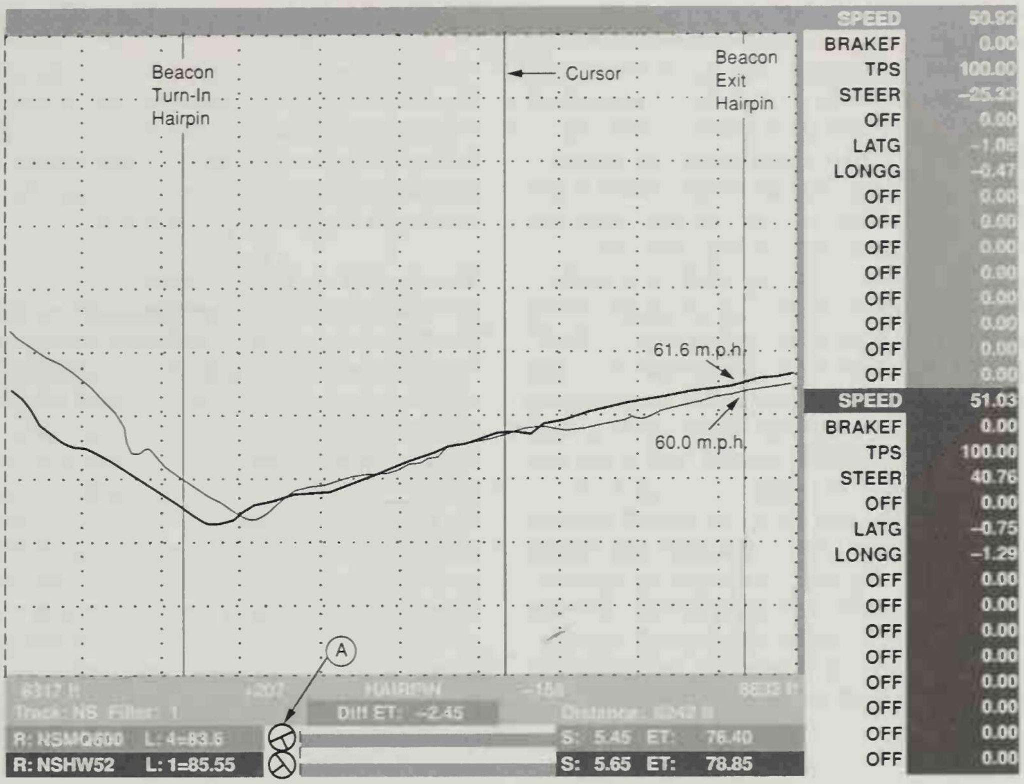

Fig. 8-11. One determinant of corner exit speed can be the amount of tire scrub or drag against the road surface, which is created by large yaw angles. The steering wheel in the right-hand Hairpin is turned 40 degrees left, indicating considerable yaw. The throttle, in this case, was never modulated, yet the exit speed is lower than the car with less yaw.

other two tenths of a second comes from. Just as you saw in Turns 7 and 9, the slower driver loses more speed than necessary: he's 7 m.p.h. down at the turn-in, 2 m.p.h. down at the minimum speed point and 1 m.p.h. off at the apex. Toward the exit of the corner, however, the slower pass makes up some of the time (.05 seconds) lost.

As you saw in Turn 9, however, overslowing the entry does allow for more aggressive acceleration; and, if you look at the speed trace, the slower driver does gain an advantage in the second part of the hairpin. At the lower entry speed, the slower driver was able to begin the acceleration through the apex of the hairpin some 30 feet earlier than the target lap, and the result was an exit speed of 61.6 m.p.h. vs. 60 m.p.h. for the target lap.

The acceleration rate is higher for a car under the absolute limit of cornering grip, but the ability to use more throttle is only part of the story. The drag from tire scrub must also be taken into account. At the yaw angle in the range of limit cornering, since the tires are sliding at an angle to the true direction of travel, they create more drag from scrubbing across the surface of the road. In a car with limited horsepower, this drag also slows the acceleration rate. Still, if the acceleration starts at a higher speed, there is usually a gain, not a loss.

In Fig. 8-11, you can see one of the reasons that the target lap speed trace falls off at the exit of the hairpin: the steering wheel for the target lap (A) is turned about 40 Degrees left in this right-hand turn— an indication of substantial oversteer. Although the driver didn't have to surrender throttle position in this instance, the additional tire scrub flattens out the speed trace, resulting in slower exit speed. Everybody, even test drivers, makes mistakes that cost lap time. On subsequent runs, McQuiston's exit speed out of The Hairpin was consistently 1.5 to 2 m.p.h. quicker.

Closing the Gap

Correcting the problems in Turns 7 through 9 and improving The Hairpin has the potential of gaining 1.35 seconds, a lap time improvement sure to bring a grin to any driver's face. The next biggest chunk, .35 seconds, could come from the Turn 3 to Turn 6 segment. Without even looking at the graphs, it would be a good bet that the lap time comes from the same faults you saw at the other corners—inefficient threshold braking and overslowing the car between the turn-in point and the minimum speed point of the corner.

If you look at the braking graph for the Turn 3 segment (see Fig. 8-12), you can see that the slower driver braked 60 feet earlier and went to a significant level of braking, 115 lbs of pedal force, initially. He modulated out to 25 lbs. again, however, to accommodate the downshift, and did most of the deceleration for the corner in the 50 lb. range. The faster pass wasn't as picture perfect as the braking for The Hairpin, but stayed in the range of 125 lbs to 90 lbs for the duration.

The throttle application in the two runs is similar. In the target lap, the throttle application begins at the same point, but it is done more aggressively. You can see by the speed trace for the target lap that there was wheelspin generated. Once the tires bit, the target lap exit speed was slightly below the slower lap but, interestingly, further along the run toward Turn 6, the target lap speed jumps ahead of the slower lap. Let's graph throttle position and see if one upshift was faster than the other, causing the target lap to pull ahead.

As you can see in Fig. 8-13, the target lap upshift takes .2 seconds from off the throttle until the throttle is at 100% again. In the slower lap, the shift takes .4 seconds and the speed advantage of 1.3 m.p.h. out of Turn 3 turns into a speed deficit of 1 m.p.h.

IMPLICATIONS OF THE DATA

The driver who participated in this computer coaching session obviously can take this information and work on the techniques that will make up the lost time. For those of you with a more general interest, you can take away some overall lessons by looking at where the lap time was in this case.

Exit Speed vs. Entry Speed

We encourage drivers to make a routine of looking at the tachometer at the exit of every corner to get an idea of whether their performance through the corner is improving or not. But if you check the exit speed numbers in these two laps, all but one corner are within a mile per hour of each other and, in some cases, the exit speed for the slower lap is even higher than for the target. So once you're close to the fastest pace, it is difficult to work on improving by checking exit speed RPM. The peek at the tach will be a tool that's most useful in the early stages of lapping a course, when you're ten seconds off the pace and working it down. When you're close, corner exit speed is often right near the maximum. You can still use the tach as a reference, but don't be surprised if you don't see much variation.

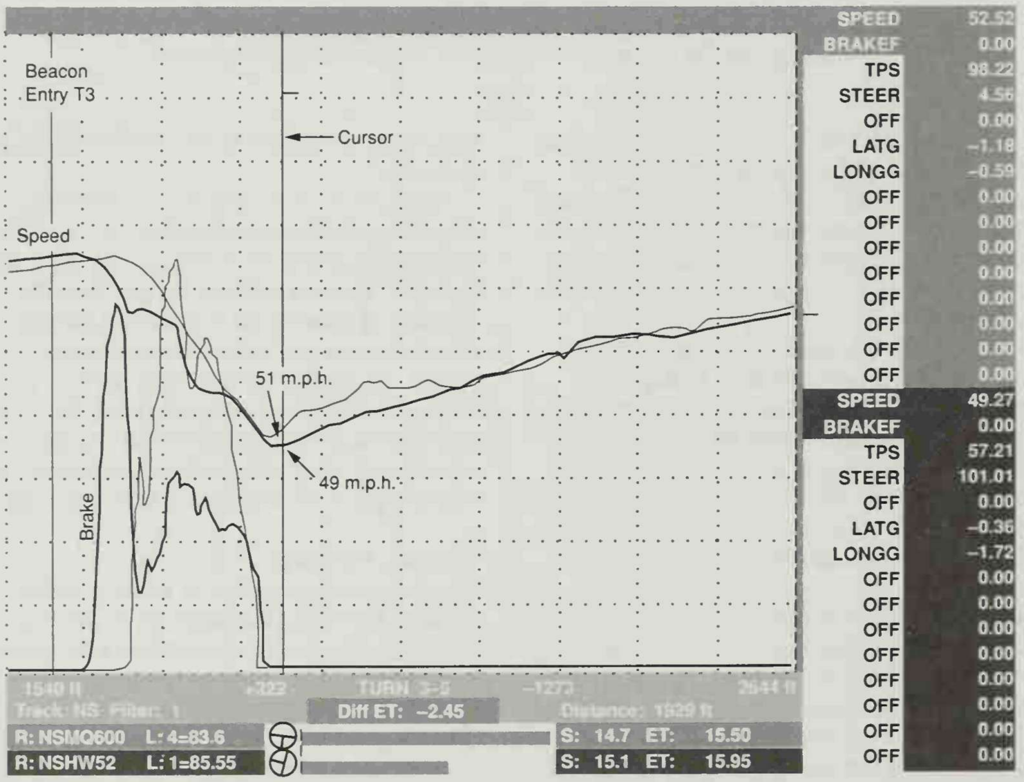

Fig. 8-12. The problem with Turn 3 is the same: poor brake level control, especially when downshifting. On the positive side, the minimum cornering speed is within 2 m.p.h. of the target run.

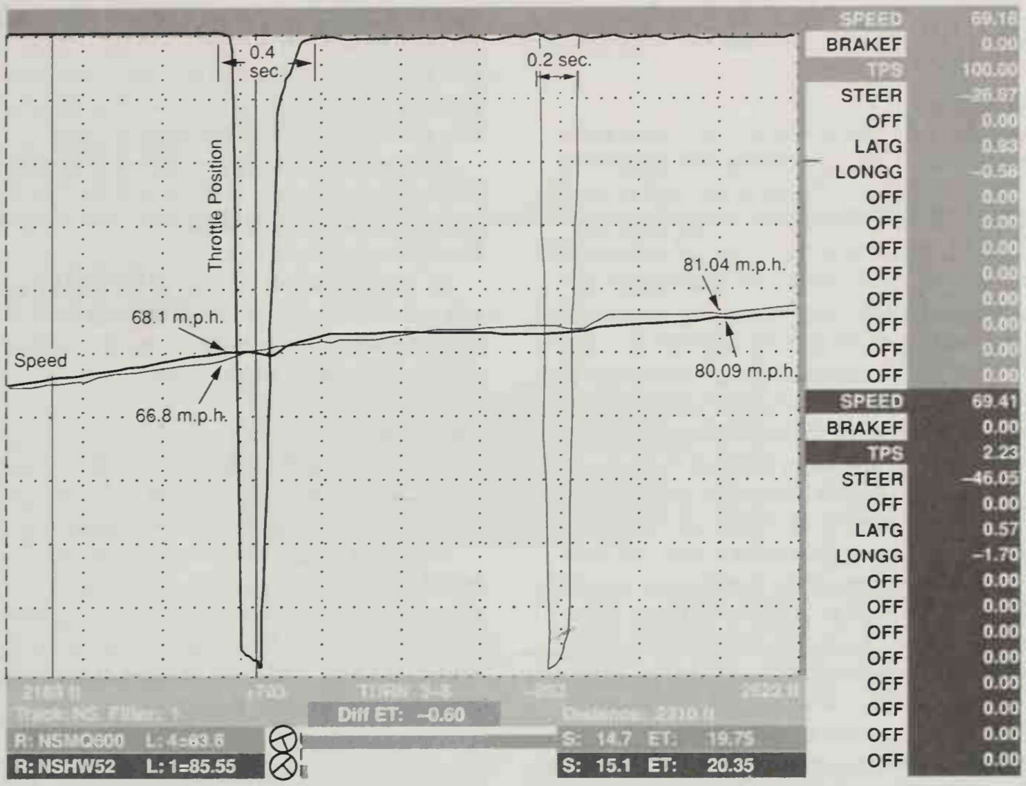

Fig. 8-13. This is a graph of throttle position and speed for the exit of Turn 3 toward the approach to Turn 6. We can see that the upshift for the target lap takes less time and, as a result, the speed trace, which was lagging slightly, bumps ahead of the slower lap.

In these comparisons, most of the time was at the corner entry. These results are typical of what we find when drivers are looking for the last percentages of improvement. Just screwing up the courage to go deeper, without developing the ability to threshold brake, will be a waste of time, but if you work on heel-and-toe technique so that you can brake effectively and do the blip at the same time without affecting the brake level, you stand a good chance of finding that lap time.

It Takes Time to Find Time

We've covered over and over again the idea of developing some perspective while behind the wheel. In these computer collections, you were looking at two drivers working within two seconds of as fast as the car will go around this racetrack. Neither driver got this good without investing the time it takes to get this close—time spent developing accuracy of line and car control skills. The driver receiving the coaching did over 50 laps on this day—laps where he was trying hard to go faster, yet he didn't have one spin or significant departure from the line. If you're spinning or missing apexes or driving sloppily, there's just no way to work on shaving the tenths of seconds here and there which eventually add up to substantial improvements of your lap time.

Beware the Red Mist

When you get frustrated with your lap time, which you will inevitably do, you have to avoid the temptation to get lathered up and just increase your aggressiveness and risk-taking everywhere around the course. In these collections, you can see that although the slower driver was almost two seconds off the fastest pace, he did quite well in numerous parts of the racetrack. Trying to go faster in the fast esses, for example, would have been a waste of time and effort—trying harder there probably would have *lost* lap time.

> *"I've often found that if I'm having trouble in a racecar, it helps to stop, get out of the car, and go away and relax for a while. I don't necessarily think about it directly—I just get away for a bit and give it a chance to slow down in [my] mind."*
>
> *—Danny Sullivan*

The reality is that significant pieces of lap time come from being just a few m.p.h. slower than the fastest driver in a few significant places. The trick is identifying those places and putting the effort there, not on parts of the circuit where you're doing fine.

Going to School

As you've seen, with a data collection system and a target time to chase, you can clearly identify where the gaps lie. But, with or without these high tech aids, self-criticism is at the core of finding lap time. Are your downshifts as good as they can be? Are you always on line? Are you at threshold braking for the high speed loss corners? Are you suiting your braking to the speed loss required by each corner? Are you working at moving throttle application points earlier? Losing less speed on the way to the apex? If you're not thinking about the specifics, finding a trick that results in a lower lap time is a crap shoot.

Short of a session with a data collection system, you can make comparisons of where on the racetrack your problems may lie by following a more skilled driver. In order to get any advantage from this, you have to get close enough so that you can see where gaps emerge and this can be tough when the better driver is pulling away at two seconds per lap. More often than not, you'll learn what you can from more than one driver as practice sessions go on.

Assuming that you can stay on line and that there are no obvious problems with braking or shifting, the likely problem areas will be what you've seen in the above data collection. In all the computer coaching we have done at the Racing School, the trend is the same. In high speed-loss situations, the ability to threshold brake is important, but more important is the speed to which you slow the car—the minimum speed point in the corner. In all corners, the speed from the turn-in to the apex is where significant time can be found.

The oddity in our example was that the slower driver had no problem with the fast stuff. That's unusual. Fast corners are more intimidating and drivers with less experience have a harder time feeling comfortable with 85 m.p.h. slides than 35 m.p.h. slides. If you can get close enough to a faster driver, you can begin to get an appreciation of what is possible—with two caveats that you must keep in mind.

1) Are you driving comparable equipment? If the faster driver has a car with more downforce or fresher tires, it might be impossible for anybody else to be as fast through the corner.

2) You might not be able to match the faster driver's talent. Let's face it: some drivers have developed the finesse and car control to master high speed slides, while others aren't there yet.

Personal Limits

Don't get caught by the "If he can do it I can do it" attitude. Get there in small bites, and if it feels too fast, maybe you've reached your personal limit.

Personal limits: it's an important idea to keep in mind. We approach driving as if everybody who gets into a racecar wants to be the fastest driver on the course. Everyone, in reality, chooses how fast they really want to go, and what combination of risks versus rewards they want to pursue.

> *"You've got to be careful making comparisons from driver to driver. Somebody says they're going to the 200 marker before braking—how fast will that car slow down compared to yours? Is he braking when he sees the 200 marker or braking at the 200 marker, or is he just not telling the truth? ...I would take any information I got from a competitor with a big pinch of salt. I'm not saying, don't talk to other drivers, but you have to be careful."*
>
> *—Danny Sullivan*